

# Data Mining and Visualization for Prediction of Multiple Diseases in Healthcare

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**Abstract:-** A vast amount of data is generated in the fields of healthcare and diagnostics, doctors have to make a direct contact with patients to determine the wounds, injuries and diseases by which the patient is affected. This paper highlights the application of classifying and predicting a specific disease by implementing the operations on medical data generated in the field of medical and healthcare. In this project an efficient multiclass Naïve Bayes algorithm is used for prediction of a particular disease by training it on a set of data before implementation. Wrong clinical decisions taken by medical practitioners can cause any harm or result in serious loss of life of a patient which is hard to afford by any hospital. To acquire a precise and cost effective treatment, technology based Data Mining Systems can be constructed to make worthy decisions. The main aim of this project is to build a basic decision support system which can determine and extract previously unseen patterns, relations and concepts related with multiple disease from a historical database records of specified multiple diseases. The proposed system can solve difficult queries for detecting a particular disease and also can assist medical practitioners to make smart clinical decisions which traditional decision support systems were not able to. The decisions taken by medical practitioners with the help of technology can result in effective and low cost treatments. There is an insufficiency of technology and analysis system and methods to discover connections, concepts and patterns in the medical data. Data mining is an engineering study of extracting previously undiscovered patterns from a selected set of data. In this paper, data mining methods namely, Naive Bayes and J48 algorithms are compared for testing their accuracy and performance on the training medical datasets. The medical datasets will be visualized by different visualization techniques like 2D/3D graphs, pie charts and other methods. The algorithms mentioned above are compared and evaluated on basis of their accuracy and time consumption factors. The algorithm which gives out high accuracy in the comparative study is selected for implementation for developing the system.

**Key Words:** Prediction, Classification, Genetic Algorithm, Naïve Bayes, Decision tree, Datasets.

## I. INTRODUCTION

Computer science is a field comprising of various technologies and techniques related to data science,

artificial intelligence, and graphical visualization. One of the new and trending technology in fields of computer and data science. Data Mining plays an important role in fields of computer science for many reasons. In this domain, Knowledge Discovery from Data also called as “KDD Process” is defined as extracting previously undiscovered knowledge from the selected target data from external sources. Data mining is a step in the KDD Process, it is an analysis technique in which previously undiscovered patterns, relations and interesting concepts are mined or extracted from the set of data. The patterns extracted are supposed to be potential, vivid and novel. Data mining can be splitted up into two parts, which is predictive and descriptive data mining. Predictive data mining includes techniques used for prediction purposes like classification and regression methods. Descriptive data mining include techniques like clustering and association rules. Ankita Dewan et.al in 2015 described the applications of data mining in fields like home automation, IoT, Artificial Intelligence, Banking and finance, Medical and Healthcare, etc. Doctors and medical people in diagnosis generate data with a less amount of hidden textual information present, this information is not used effectively and efficiently for prediction of disease[1]. To avoid this issue, the unused data is converted into a textual and tabular dataset for modelling of data using various data mining methods and algorithms. Because of inefficient results and issues that were not taken into consideration at the time of treatment, a person can die. There is a serious need for practitioners to analyse and shape and modelled data and identify the disease cause of a patient and predict the disease before it occurs. The predictions and classifications techniques in data mining help discover new trends and patterns in medical data in order to improve their health[1]. The issues of data mining represent great challenges for networking and distributed computing. Various scientific researchers had suggested several methods to deal with the problems of data mining. Data mining have achieved excellence in fields of web technology and interfacing by deploying new concepts and linking them with web technology to secure human lives. Adam Sadilek et.al in their article which was published in NewYork described the use and linking of data mining to web technology. They worked on preventing food illness, which is also the damage causing disease in the United States[2]. A web interface named “nEmesis” was developed by these researchers and it used to mined social media content and help those patterns to prevent food illness by showing the results and food bitmaps in the corners of U.S. Jyoti Soni et.al in their article paper proved that the operations on the medical datasets were conducted using data mining algorithms like K-NN, Decision trees, and Naive Bayes and the results proved that Naive Bayes

technique outclassed the accuracy test[3]. The classification and prediction techniques can be improved by concatenating association rules with it to find out the frequently used elements. The algorithms such as dynamic Apriori algorithm and FP-Growth are used to mine frequent itemsets from a database. In the previous research works, the implementation and the accuracy testing of algorithms such as decision tree, back propagation (Neural Networks), Naïve Bayes and K-NN was carried out in which the Neural Network's back propagation model achieved the highest accuracy of them all.

## II. RELATED WORK

Various data mining algorithms and techniques have been used for study and analysis of various diseases like cardiac arrhythmia, Hepatitis, Diabetes Cancer etc. Recent survey shows that heart disease is one of the biggest cause of death in the countries like UK, Canada, France and Singapore. Various classification models like Decision Tree, Naïve Bayes and Neural Networks, KNN have been used to diagnose the presence diseases in patients. Pattern recognition and data mining methods have been used previously by the practitioners and researchers for prediction purpose in the field of diagnosis and healthcare. The experiments were carried out using data mining algorithms like Naïve Bayes, Decision Tree, K-NN and Neural Network out of which Naïve Bayes gave better results. An experiment carried out by the researchers using neural networks and hybrid intelligent algorithm on a dataset produced a model and the results shows that the combined hybrid and intelligent technique improved accuracy of the prediction. The technique which is used to find out the interesting patterns and relationship between two items is called as association rule. The two items in a transactional database can share a dependent relationship. Abhishek Taneja in his paper mentioned that the algorithms with search limitations and constraints are also introduced to reduce the count of association rules and also for validation purpose[4]. T.Revathi et.al in their IJSR's vol- 4[2015] explained about the implementation of classifier models and their accuracy and latency analysis[5]. Practitioners used popular data mining techniques such as decision tree, KNN naive bayes to develop a model which can be used for prediction and analysis of various diseases. The authors in their survey paper described the previously evaluated results of algorithm comparison and prediction accuracy on the medical datasets. The algorithms used for comparative study were Naïve Bayes, decision tree, K-NN and back propagation method[6]. Classification is a supervised learning method used for elaborating the famous classes of data. Classifiers like Naïve Bayes, Decision Tree, and also clustering have been used to diagnose and predict the heart disease in patient. The challenge of positioning and together grouping of similar elements was sorted by Hui Xiong et.al in their article based on clustering in 2004[7]. There are various applications of classification such as medical diagnosis, student track-up system and also can be used in field of biology for genetic identification. Nuzhat F. Shaikh et.al in their paper described the implementation of neural networks and artificial intelligence techniques to improve the accuracy of iris recognition system and making it user convenient[8].

## III. MOTIVATION

In the healthcare organizations and medical diagnosis, artificial intelligence and machine learning techniques and algorithms can be implemented to treat some hazardous diseases. Besides all the other diseases, cardiac arrhythmia, arthritis, Ischemic Heart disease, Hepatitis, Marasmus are responsible to take more lives than any other diseases such as cancer and HIV. Thus, there is a need to predict these diseases in real time to prevent future hazards and to avoid further issues and problems. A tabular dataset is constructed by encapsulating all of the medical digital data and is used for further experiments of prediction analysis. The selected target algorithms are weighed by considering factors like classification and latency accuracy and the algorithm that gives out highest accuracy is selected to classify the previously unknown and unseen data tuple with examples according to its class.

### A. Existing System

In the previous years, researchers and practitioners in the field of medical and healthcare have achieved success in predicting a specific disease using data mining algorithms and techniques. It was stated that the use of classification algorithms like decision tree and back propagation method to predict heart disease using training datasets which were downloaded from the official UCI repository which is deployed on the internet. One of the drawbacks of the previous and existing systems are that they were related to classification algorithms and medical datasets were implemented by considering only one medical dataset only.

### B. Proposed Work

The abstract and sections mentioned above proposes an advanced and efficient prediction of multiple diseases based on past historical training data. The ideal strategy is to analyse and test various data-mining algorithms and to implement the algorithm which gives out highest degree of accuracy. The Datasets selected for implementation purpose contains more than 20 medical related attributes for each disease and maximum upto 500 algorithms training data examples. The medical datasets selected for the project are heart disease, breast cancer, Arthritis and diabetes. The algorithm used to implement in this project is Naive Bayes, which is selected by evaluating the predictive accuracy and latency analysis results. The project also consists of a visualization module in which the medical datasets are displayed in a diagrammatic representation using different data visualization techniques for user convenience and better understanding.

## IV. DATA MINING TECHNIQUES

### A. Data Mining Algorithms

#### 1. Decision tree Induction:

A decision tree is a data mining structure which resembles a tree like diagram consisting of number of nodes and edges. The nodes represent the test on the attributes and the edges represents the outcome of the test on the attribute. The IF-THEN rules are used to build up rules on any decision tree. The learning of decision tree is called as decision tree induction. The algorithms used for decision tree construction are C4.5, CHAID, CART, and the most commonly used algorithm is ID3( Interactive Dichotomiser 3). The decision

tree algorithm has the best advantage of all the time, that is it can handle numerical as well as categorical data without any interference. Monika Gandhi et.al in their article which was published in INDIACOM conference in 2015 gave a basic overview about decision tree induction, this data mining algorithm can handle both numerical data and categorical data[9]. The most commonly used algorithm to build a decision tree is ID3(Interactive dichotomized 3) . A decision tree is a structure consisting of leaf, roots and branches. The leaves in the tree represents the class label, where the rules constructed upon the tree are used to find out the class label of the previously unseen examples.

- **J48 Decision Tree**

Decision Tree Algorithm is to find out the way the attributes-vector behaves for a number of instances. The classes for the newly generated instances are being found on the basis of the training instances. This algorithm can be used to construct some rules that can be used for the prediction of the class label of the testing data. The critical distribution of the data is easily understandable with the help of tree classification algorithm. J48 is an extension of ID3. Other features of J48 algorithm are that it supports tree pruning, can handle missing values, and also gives out efficient output in prediction analysis In WEKA, J48 is an open source Java implementation of the C4.5 algorithm and developed by WEKA's development team. According to the results when J48 and Naïve Bayes were compared for analysis purpose, both the algorithms performed efficiently, but as Naïve Bayes works good with large number of data instances, it outclassed the performance test.

- **Naïve Bayes Classifier**

Naive Bayes is a technique for constructing classifiers. The classifier is used to assign class labels to various problem instances using the concepts of statistics and calculations. It is model which requires only a finite number of training data to evaluate the fields required for predictive mining. Baye's theorem serves as the base for the naïve Bayes algorithm, in which it explains the relationship between conditional or dependent probability and unconditional or independent probability. The equation(1) given below describes and outlines the entire Baye's theorem and the relationships between the probabilities. Bayes theorem was invented to deduce the links and relationships between the probabilities. There are basically two types of probabilities, one is the conditional and other is unconditional probability. The probability which is ruled out before obtaining the evidence is called as unconditional or prior probability. The probability which comes into play after obtaining the evidence is called as conditional or posterior probability. In the equation mentioned below, P(A) and P(B) are the prior or unconditional probabilities, and P(A|B) and P(B|A) are the conditional probabilities where the first event depends on the occurring of the second event.

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \quad \text{Eq(1)}$$

### B. Data Mining Platforms

Various open source data mining platforms are available online or on the internet markets. These applications are free, licensed, and small in size. WEKA, Rapid Miner,

Bayesian Lab, etc are some of the open source data mining platforms available on the internet markets. The platform used in this project is “WEKA 3.7”, which is also the latest version of WEKA tool software available presently. The sources from the internet says that. WEKA as the name sounds stands for Waikato Environment for Knowledge analysis, this software was coded in java in its early stages and later on was evolved for operating on datasets. WEKA contains a plethora of graphical visualization methods and algorithms for data analysis and prediction purposes[10]. WEKA supports many data mining tasks like pre-processing, association, clustering, and classification. WEKA tab consists of user interfacing options such as explorer, knowledge utility, CLI and Workbench options. The explorer option in the tab is used for selection of any dataset which is in .arff form and to perform operations on it such as clustering, prediction, association and visualization. WEKA provides a connectivity bridge to SQL databases using Java Database Connectivity over ODBC (Open database connectivity) and can process the result returned by a database query given by the user. Areas like sequence modelling and others are not covered under WEKA Distribution. Some Snapshots of WEKA processing the arthritis dataset are been described below.

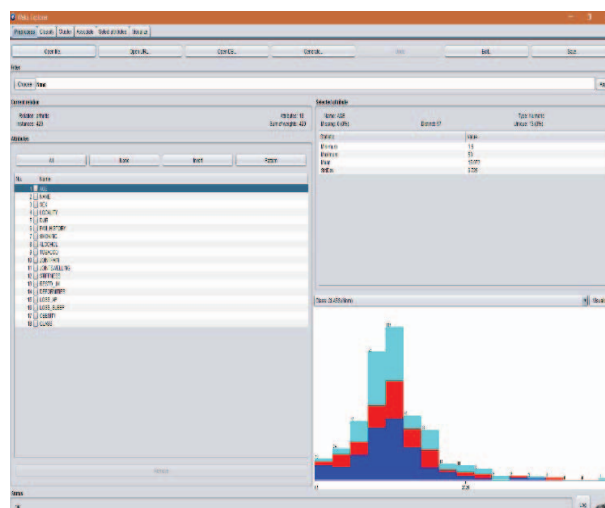


Fig 1: Explorer View of attributes of dataset using WEKA

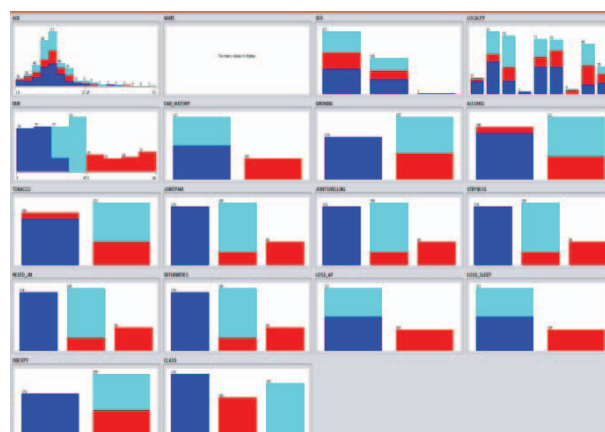


Fig 2: Visualization of all the attributes of the dataset in WEKA

## V. DATA VISUALIZATION

Data in tremendous amount is very difficult to understand and interpret. The big data can be represented in form of interesting shapes in different sizes using lines and dots. The graphical visualization of this data helps to display the concepts in simple way and lead to ease of understanding. Visual data mining helps to deal with the huge data and flood of information, integrating the human in data analytics process. It allows user to acquire the insights of the data and derive conclusions by interacting with the data. Prof. Pier Luca et.al in their article of "Visualization in Data Mining" mentioned the approaches of data visualization in brief[11]. The approaches in data visualization are described below:

1. Presentation: Very first stage, in which data is presented using presentation methods.
2. Confirmative Analysis: The hypothesis of the data is either confirmed or rejected in this phase.
3. Explorative Analysis: The tests are carried on the data without any analysis.

### A. Data Visualization Techniques

Data visualization is a technique of presenting the data in graphical format. Graphical visualization usually deals with nodes and links in which the nodes or the vertices represent the attributes and the links or the edges represent the connection between the attributes. Graphical or graph visualization is the simplest type of visualization and which provide a better way of exploring and understanding data. Some techniques of the chosen visualization techniques which are also used in the project are enlisted below:

a) Line Graphs      b) Pie charts      c) Bar graphs

- 2D/3D Line Graphs

The data is visualized in form of lines and dots in multiple dimensions. The graph can have multiple axes and the readings can be plotted at the co-ordinates on the plane. The figure given below depicts the details about the line graph.

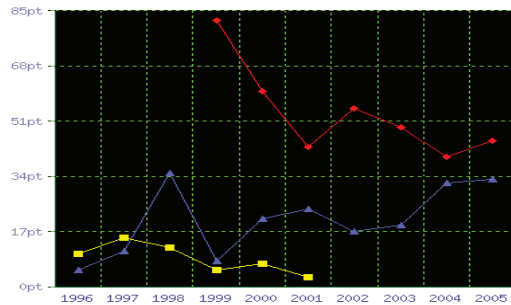


Fig 3: 2D Line graph

- Bar Graphs

A bar graph is mathematical representation of data. Bar graph shows comparison between different categories of the data. The dependency and correlation between the attributes is represented in form of solid bar shapes on the graph. If the descriptions are on the horizontal axis, the bars will be oriented vertically, and if on the vertical axis

then can be oriented horizontally. Various types of bar graphs are simple bar graph, vertical and horizontal bar graph, range bar graph and histogram.

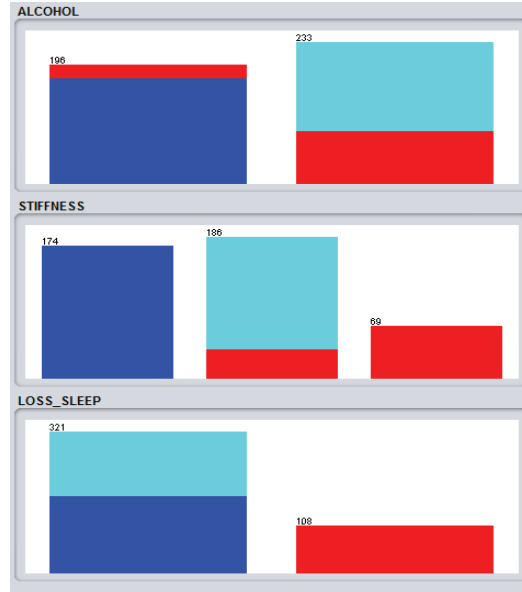


Fig 4: Bar graph visualization of Medical dataset

The above given figure describes the bar graph visualization of the attributes such as Alcohol consumption, Early morning stiffness and Loss of sleep. The colours red, blue, and faint green indicate the classes such as negative, positive and may occur in future.

- Pie Charts

A pie chart is used to describe a big chunk of data into small slices of data where each slice indicates the amount of elements in the category of the dataset. Mostly, this type of visualization is used in statistics and mathematical computations.

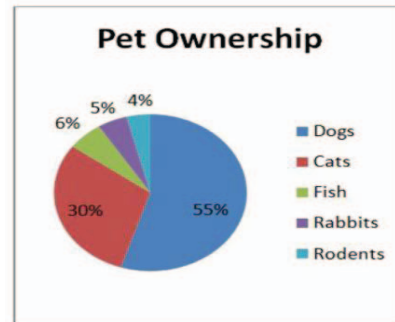


Fig 5: Pie- chart visualization Technique

As shown in fig 5, the pie chart shows different categories such as sdogs, cats, fish, rabbits, etc in a mathematical percentage count.

- Advantages of Data Visualization:

Data visualization has many advantages, few of them are listed below:



1. Ease of understanding and exploration of data
2. No complex statistical and mathematical calculations are required for visualization.
3. A better way of knowledge filtering and knowledge representation.

## VI. EXPERIMENTAL RESULTS

As mentioned in the above section, the algorithms used for comparison were Naïve Bayes and Decision tree's J48. These two algorithms were tested on datasets built in .arff and .csv formats. Datasets of few diseases were downloaded from the UCI repository whereas few of them were been constructed by consulting doctors and researching about the disease. The datasets of disease selected were Pima Indian diabetes, breast cancer, heart disease and arthritis. Every dataset consisted of about 100 to 700 data instances and 14-19 attributes[12]. The algorithms selected for comparison were judged on the basis of accuracy and time taken for prediction of class labels.

Table 1: Accuracy analysis of algorithms on the datasets

| Algorithm   | Accuracy on Diabetes dataset | Accuracy on Breast cancer dataset | Accuracy on Heart Disease dataset | Accuracy on Arthritis dataset |
|-------------|------------------------------|-----------------------------------|-----------------------------------|-------------------------------|
| Naïve Bayes | 76.30%                       | 71.45%                            | 83.49%                            | 100%                          |
| J48         | 73.82%                       | 75.52                             | 77.55%                            | 98.12%                        |

The clinical data related to diseases like diabetes, Breast cancer, Heart disease and arthritis were obtained manually or by downloading it from the UCI repository. The algorithms like Naïve Bayes and J48 were tested on these datasets on the famous data mining platform called "WEKA". The training sets of the data were dismantled into 10-cross validation and by using stratification techniques. By considering observations shown in the above table 1, we can conclude that the Naive Bayes outclasses the comparative test with J48 and excels the highest percentage of accuracy on all the datasets taken into consideration.

Table 2: Latency Analysis of algorithms on the datasets

| Algorithm   | Time Taken to build Diabetes dataset | Time Taken to build Breast cancer dataset | Time Taken to build Disease dataset | Time Taken to build Arthritis dataset |
|-------------|--------------------------------------|-------------------------------------------|-------------------------------------|---------------------------------------|
| Naïve Bayes | 0.01 seconds                         | 0 seconds                                 | 0 seconds                           | 0 seconds                             |
| J48         | 0.02 seconds                         | 0.4 seconds                               | 0.01 seconds                        | 0 seconds                             |

As shown in the above table 2, the performance study of algorithms on the basis of time taken to build the dataset or the latency duration was also conducted in the performance analysis process. WEKA was the data mining platform used, where Multiclass Naive Bayes and Decision tree's J48 were used for implementing on the training datasets. Naïve Bayes outperformed this test by giving out a minimum latency of 0 seconds to 0.01 seconds on each target dataset. While, J48 resulted in giving a building time domain from 0 seconds to 4 seconds on the target datasets. By considering all the performance results and tests conducted on the datasets, Naïve Bayes algorithm was selected for implementation in further project work and analysis.

### A. Description of Datasets

The datasets used in the project consists of attributes and data instances in large amount. The count of attributes and training data instances in each dataset is being described below:

1. Pima Indian Diabetes Dataset  
Attributes: 9  
Data Instances: 768
2. Breast Cancer Dataset  
Attributes: 10  
Data Instances: 286
3. Heart Disease Dataset  
Attributes: 14  
Data Instances: 270
4. Arthritis Dataset  
Attributes: 18  
Data Instances: 429

Table 3: Attributes of target datasets

| Diabetes Dataset                                                                                                                         | Breast Cancer Dataset                                                                                                                                                                           | Heart Disease Dataset                                                                                                                                                                                       | Arthritis Dataset                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Age<br>2. Count of Pregnancy<br>3. Diastolic Blood Pressure<br>4. Obesity (Skin Thickness)<br>5. Body Mass Index<br>6. Family History | 1. Age<br>2. Menopause Detail<br>3. Tumor Size<br>4. INV- Nodes<br>5. Cap-Nodes<br>6. Degree of malignancy(Obesity)<br>7. Breast side<br>8. Breast Quadrant<br>9. Irradiation treatment(yes/no) | 1. Age<br>2. Sex<br>3. Chest Pain Type<br>4. Blood pressure<br>5. Serum Cholestora<br>6. Fasting blood sugar<br>7. Thalach Reading<br>8. Oldpeak induction<br>9. Slope Readings<br>10. No. of major vessels | 1. Age<br>2. Sex<br>3. Duration of disease(days)<br>4. Family History<br>5. Smoking habit<br>6. Alcohol habit<br>7. Tobacco Habit<br>8. Joint pain level<br>9. Joint Swelling level<br>10. Early Morning Stiffness<br>11. Loss of appetite<br>12. Loss of Sleep<br>13. Obesity |

Attributes such as name, age and sex were common in all the target datasets. Other attributes which are considered for prediction are family history and obesity. The end patient users are supposed to enter the few parameters at the front end user interface. The classification algorithm which is running at the back end on the datasets compare the testing data, also called as previously unseen data with the training data, also called as the data in the dataset. The input data whose properties match with the training data is returned as class label of that instance. In critical disease dataset such as breast cancer, attributes such as Menopause details, tumour size, breast size and side plays a major role in the classification of disease. Arthritis, is a long standing chronic

disease which causes many disabilities. Arthritis can occur at any age, usually seen in young women. There are many types of arthritis that can occur in small children also. If not properly treated at the early stages it can cause deformities in the early ages. The major factors or parameters which are seen as the symptoms of arthritis are joint pain, joint swelling, early morning stiffness, obesity, alcohol and smoking habits, loss of sleep and loss of appetite. Then input for symptoms such as joint pain and swelling level is taken as high, medium and low from the end users. For obesity, alcohol and smoking habit input is taken as yes or no. The final class label of the dataset can be either of the three, that is positive, negative or can occur in future.

## VII. CONCLUSION

The current advancement, development and improvement in the data mining algorithms have assured an ease in obtaining insights and precise prediction results. The paper enlightens all the challenges, issues and obstacles faced by medical practitioners while determining the disease of a patient without considering the medical data of the patient. The conclusion derived from the evaluation results of J48 and Naïve Bayes, it was observed that Naïve Bayes results in best accuracy of prediction and even outclasses J48 in the latency analysis on the datasets. The proposed work is efficient enough to build an expert decision support system which can be used by medical practitioners for enhanced diagnosis and also can be used by patient users for prediction of disease. All the confidential medical data is gathered from the hospitals and UCI repository and algorithms are applied on them for the best performance analysis. The paper also outlines the techniques and types of data visualization techniques which are used in this project system and implemented on the medical datasets for ease of data understanding and exploration of data. The system can also be deployed on android and web platforms for user convenience by porting it by various conversion applications

## VIII. REFERENCES

- 1] Ankita Dewan, Meghna Sharma , "Prediction of Heart Disease Using a Hybrid Technique in Data Mining Classification" , 2015 2ndInternationalConference on Computing for Sustainable Global Development (INDIACom), IEEE 2015.
- 2] Adam Sadilek, Henry Kautz, Lauren DiPrete, Brian Labus, Eric Portman<sup>1</sup>, Jack Teitel<sup>1</sup>, and Vincent Silenzio, " Deploying nEmesis: Preventing Food borne Illnessby Data Mining Social Media",NewYork.
- 3] Jyoti Soni, Ujma Ansari, Dipesh Sharma, Sunita Soni , "Predictive Data Mining for Medical Diagnosis: An Overview of Heart Disease Prediction" , International Journal of Computer Applications (0975 8887) Volume 17 No.8, March 2011.
- 4] Abhishek Taneja. "Prediction of heart diseases using data mining techniques". Oriental Journal of computer science and technology. December 2013. Vol. 6, No. (4): Pgs. 457-466.
- 5] T. Revathi<sup>1</sup>, S. Jeevitha, "Comparative Study on Heart Disease Prediction System Using Data Mining Techniques ",International Journal of Science and Research (IJSR) ISSN (Online): 2319-7064 Index Copernicus Value (2013).
- 6] Ajinkya Kunjir, Harshal Sawant, Nuzhat Shaikh," A Survey on Prediction of Multiple Diseases and Performance Analysis using Data Mining and Visualization Techniques", International Journal of Computer Applications (0975 – 8887) Volume 155 – No 1, December 2011.
- 7] HICAP:Hierarchial Clustering with Pattern Preservation (2004). Hui Xiong, Michael Steinbach, Pang-Ning Tan, and Vipin Kumar, In Proc. of the Fourth SIAM International Conf. on Data Mining (SDM'04), Florida, USA, 2004.
- 8] Nuzhat F. Shaikh, Dharmpal D. Doye, "Improving the Accuracy of Iris Recognition System using Neural Network and Particle Swarm Optimization" International Journal of Computer Applications (0975 – 8887)Volume 79 – No3, October 2013
- 9] Monika Gandhi, Dr..Shailendra Narayan Singh ,"Predictions in Heart Disease Using Techniques of Data Mining" , 2015 1st International Conference on Futuristic trend in Computational Analysis and Knowledge Management (ABLAZE-2015), IEEE 2015.
- 10] [www.cs.waikato.ac.nz/ml/weka/](http://www.cs.waikato.ac.nz/ml/weka/)
- 11] Prof. Pier Luca Lanzi, " Visualization Techniques in Data Mining".
- 12] [archive.ics.uci.edu/ml](http://archive.ics.uci.edu/ml)